Measuring Financial Connectedness among Emerging Equity Markets: Diebold–Yilmaz, NETS and FNETS Compared*

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Summary of the Base Paper and of the Network Estimators

Diebold and Yılmaz (2014) define connectedness through the H-step generalized forecast-error variance decomposition (GFEVD) of an approximating VAR. The share of the forecast-error variance of variable i due to shocks in variable j is read as a weighted, directed link $j \to i$; row sums of the off-diagonal entries give how much each node receives ("from"), column sums how much it transmits ("to"), and the cross-sectional average is total connectedness. Using the generalized decomposition of Koop et al. (1996) and Pesaran and Shin (1998) makes the network invariant to the ordering of the variables. Rolling-window estimates track how connectedness changes over time.

NETS (Barigozzi and Brownlees, 2019) estimates the network directly as a sparse object: a lasso-type estimator of a VAR delivers a *Granger* network (lagged dependence), the sparse concentration matrix of the VAR innovations delivers a *contemporaneous* partial-correlation network, and their combination delivers a *long-run* partial-correlation network.

FNETS (Barigozzi et al., 2024) first removes a (possibly dynamic) common factor structure, estimated by dynamic principal components in a generalized dynamic factor model (Forni et al., 2000), and then estimates a sparse VAR and sparse (long-run) partial correlations for the idiosyncratic component. It therefore separates connectedness due to pervasive common shocks from idiosyncratic, network-type interdependence.

[To be completed: professor's papers on the idiosyncratic component, to be added]

1 Motivation

[To be completed: Why measuring connectedness among emerging equity markets matters (contagion, portfolio diversification, policy spillovers). Argentina is treated as one case among the 17 markets (its final role, decision 6 of paper structure v2, is still to be defined by the group). Suggested anchors from the data: the Argentine primary-election shock of August 2019 (largest weekly ARGT drop, -28.4% log return) and the March 2020 crash.]

2 Introduction

[To be completed: Research question (working version): how does measured connectedness among emerging equity markets change with the estimation method (Diebold–Yilmaz, sparse VAR, NETS, FNETS), and what share of it is a global common factor versus idiosyncratic? State hypotheses and the contribution; summarize the main results once Section 5 is written.]

3 Data

Universe and sample. We use US-listed, USD-denominated country exchange-traded funds (ETFs) as tradable proxies for 17 emerging equity markets: Argentina (ARGT), Chile (ECH), Brazil (EWZ), Mexico (EWW) and Peru (EPU) in Latin America; South Korea (EWY), Taiwan (EWT), India (EPI), China (FXI), Thailand (THD), Indonesia (EIDO), Malaysia (EWM), Vietnam (VNM) and the Philippines (EPHE) in Asia; South Africa (EZA), Turkey (TUR) and Poland (EPOL) in EMEA. The S&P 500 ETF (SPY) is included as a global reference node. Funds were selected if their price history starts by 3 March 2011 (7-day tolerance), at most 5% of weeks are missing, and average daily traded volume is at least USD 3 million. Colombian funds (ICOL, GXG), INDA and MCHI did not pass the criteria; India and China are represented by EPI and FXI. The common sample runs from 11 March 2011 to 23 October 2026, i.e. 816 weeks, with no missing values.

Returns and frequency. Prices are dividend- and split-adjusted, so returns are total returns. We use weekly Friday-to-Friday log returns. The weekly frequency follows the course indication that weekly returns retain some forecastability that is absent at daily frequency; it also mitigates non-synchronous trading between Asian and EMEA local markets and their US-listed ETFs. Unlike Diebold and Yılmaz (2014), who study (realized) volatilities, we study returns: no daily data are available to build realized weekly volatility, and returns are the object of the forecastability argument. Returns are not devoid of cross-market predictive content: in a full-sample VAR(1) on the 18 series, 59 of the 324 slope coefficients (18.2%) are significant at the 5% level, against 5% expected by chance. Our networks therefore describe the transmission of return shocks, not of risk.

Descriptive statistics. Table 1 reports summary statistics. All series are strongly leptokurtic and mostly negatively skewed; an augmented Dickey–Fuller test rejects a unit root for every series. Five weekly log returns exceed 25% in absolute value (ARGT in the week of the August 2019 primary elections and four observations in March 2020); they are genuine market moves and are kept. The March 2020 crash dominates the tails: of the ten largest weekly drops pooled across the 17 emerging markets, eight occur in the weeks ending 13 and 20 March 2020, and 14 of the 17 markets record their worst week in March 2020.

Table 1: Descriptive statistics of weekly log returns, 2011-03-11 to 2026-10-23 (816 weeks). Kurtosis is not in excess.

Ticker	Country	Region	Start	Mean (%)	Std (%)	Min (%)	Max (%)	Skew.	Kurt.
ARGT	Argentina	Latin America	2011-03-11	0.15	4.18	-28.4	24.0	-0.51	8.9
ECH	Chile	Latin America	2011-03-11	-0.03	3.56	-21.2	16.2	-0.36	5.6
EWZ	Brazil	Latin America	2011-03-11	0.01	4.26	-26.5	22.2	-0.18	6.4
EWW	Mexico	Latin America	2011-03-11	0.07	3.29	-24.0	10.3	-0.74	7.8
EPU	Peru	Latin America	2011-03-11	0.12	3.35	-19.3	13.3	-0.39	5.9
EWY	South Korea	Asia	2011-03-11	0.14	3.66	-20.3	16.0	-0.40	6.6
EWT	Taiwan	Asia	2011-03-11	0.23	2.85	-12.6	11.2	-0.43	4.4
EPI	India	Asia	2011-03-11	0.09	2.90	-18.8	13.3	-0.35	6.3
FXI	China	Asia	2011-03-11	0.01	3.47	-12.6	17.0	0.03	4.5
THD	Thailand	Asia	2011-03-11	0.07	2.91	-23.5	9.6	-0.85	9.0
EIDO	Indonesia	Asia	2011-03-11	-0.08	3.68	-33.6	20.7	-1.25	14.6
EWM	Malaysia	Asia	2011-03-11	0.01	2.25	-12.2	10.9	-0.22	6.5
VNM	Vietnam	Asia	2011-03-11	-0.01	3.15	-19.3	11.9	-0.35	5.9
EPHE	Philippines	Asia	2011-03-11	0.02	2.92	-19.3	14.4	-0.57	7.4
EZA	South Africa	EMEA	2011-03-11	0.07	4.06	-26.9	14.1	-0.57	6.5
TUR	Turkey	EMEA	2011-03-11	-0.03	4.60	-23.0	20.0	-0.63	6.1
EPOL	Poland	EMEA	2011-03-11	0.08	3.66	-25.1	15.2	-0.58	6.5
SPY	United States	Global	2011-03-11	0.25	2.23	-15.7	11.4	-0.74	9.3

Limitations. (i) ETF returns embed exchange-rate movements against the USD and are subject to non-synchronous trading. (ii) ARGT changed its underlying index in August 2014 (from a Solactive Argentina index to the MSCI All Argentina 25/50). (iii) FXI tracks only the 50 largest Chinese firms (FTSE China 50) and EPI is the earnings-weighted WisdomTree India fund, so neither is a broad market index. (iv) Colombia is absent. (v) Liquidity requirements and the common start date exclude younger funds. (vi) Without daily data, realized-volatility connectedness cannot be computed.

4 Methods

Let $y_t \in \mathbb{R}^N$ be the vector of weekly returns ($N = 18$). All methods are estimated on rolling windows of $w = 104$ weeks; each estimate is dated at the last week of its window. For the comparison, all methods use the same 179 windows (step of four weeks, first window ending 1 March 2013).

4.1 Diebold–Yilmaz connectedness

We fit a VAR(p), $y_t = c + \sum_{k=1}^{p} A_k y_{t-k} + \varepsilon_t$, $\mathbb{E}[\varepsilon_t \varepsilon_t'] = \Sigma$, by OLS, with $p \in \{1, 2\}$ selected by BIC in each window. The cap $p \leq 2$ is an adaptation: with $N = 18$ and about 100 observations per window, $p = 3$ would mean 55 regressors per equation. With MA coefficients Ψ_h ($\Psi_0 = I$), the H-step generalized variance share (Pesaran and Shin, 1998) is

$$\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' \Psi_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i' \Psi_h \Sigma \Psi_h' e_i}, \qquad \tilde{\theta}_{ij} = \theta_{ij} \Big/ \sum_j \theta_{ij}, \tag{1}$$

which is invariant to the ordering of the variables. Following Diebold and Yılmaz (2014), $C_i^{\text{from}} = 100 \sum_{j \neq i} \tilde{\theta}_{ij}$, $C_j^{\text{to}} = 100 \sum_{i \neq j} \tilde{\theta}_{ij}$, net connectedness $C_i^{\text{net}} = C_i^{\text{to}} - C_i^{\text{from}}$, pairwise net $100(\tilde{\theta}_{ji} - \tilde{\theta}_{ij})$, and total connectedness $C = \frac{100}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$. The baseline horizon is $H = 4$ weeks. The base paper uses daily data, 100-day windows and $H = 12$ days; we keep the logic (a horizon of about one month, a window long enough for a stable VAR) on the weekly scale.

4.2 Sparse VAR (lasso)

As a bridge between the dense DY VAR and the sparse network estimators, we estimate each VAR equation by the lasso (Tibshirani, 1996) on lags $1, \ldots, 4$ (standardized regressors, unpenalized intercept), so that the penalty performs lag selection. The penalty is chosen per equation by time-series cross-validation (five expanding folds, test blocks always after the training block; random K-fold would leak future information), over 40 log-spaced values; the extended BIC (Chen and Chen, 2008) is an alternative. The same GFEVD and connectedness measures are computed on the lasso VAR, so differences with DY isolate the effect of regularization.

4.3 NETS

NETS (Barigozzi and Brownlees, 2019) jointly estimates a sparse VAR(1) and a sparse concentration matrix $K = \Sigma^{-1}$ of the innovations by adaptive lasso (Zou, 2006; Peng et al., 2009). Since the `nets` package requires the penalty to be supplied, we select it by BIC over 15 log-spaced values. The Granger network is given by the non-zero entries of A_1, the contemporaneous network by the partial correlations $-K_{ij}/\sqrt{K_{ii}K_{jj}}$, and the long-run network by the partial correlations of $K_{LR} = (I - A(1))' K (I - A(1))$. Each window is standardized before estimation.

4.4 FNETS

FNETS (Barigozzi et al., 2024; Owens et al., 2024) assumes $y_t = \chi_t + \xi_t$, where the common component χ_t is driven by q dynamic factors and the idiosyncratic component ξ_t follows a sparse VAR. We use the unrestricted (generalized dynamic) factor model, select q with the Hallin and Liška (2007) information criterion (criterion IC5, the package default), estimate the idiosyncratic VAR(1) by lasso with the penalty chosen by the package's BIC, and estimate contemporaneous and long-run partial correlations of ξ_t with the package's CLIME-type estimator (Cai et al., 2016) and adaptive thresholding. The share of variance explained by the common component is $\text{var}(\chi_{it})/\text{var}(y_{it})$. Applying the GFEVD to the idiosyncratic VAR yields *idiosyncratic connectedness*; comparing it with DY measures how much connectedness is due to the common factor. With $N = 18$ and $T = 104$ the estimated idiosyncratic innovation covariance is often not positive definite; we project it onto the positive-definite cone by clipping eigenvalues at 1% of the mean absolute eigenvalue, and clip partial correlations to $[-1, 1]$ (the support of the network is unchanged).

4.5 Comparable measures and comparison criteria

Every method is mapped to the same objects: a VAR on which the same GFEVD connectedness is computed; a directed Granger network (sum over lags of the absolute coefficients); and contemporaneous and long-run partial-correlation networks (for DY and the lasso VAR, implied by the estimated VAR). We compare methods on three criteria:

1. *Density*: share of non-zero links in each network; for the dense GFEVD table, share of pairwise shares above $1/N$; and total connectedness.

2. *Agreement*: correlation of total and net connectedness series across methods; per-window Spearman correlation between the off-diagonal entries of two methods' matrices; Jaccard similarity of their strongest 10% of links.

3. *Stability*: variability of total connectedness, and Spearman correlation and top-link Jaccard similarity between consecutive windows' matrices.

Finally, the share of DY connectedness that disappears once the common factor is removed is $1 - C^{\text{FNETS}}/C^{\text{DY}}$, averaged over windows.

4.6 Robustness

We rerun all methods with windows of 78 and 130 weeks, horizons of 2 and 8 weeks, without SPY, and excluding 2020. For the last, every rolling window containing any week of 2020 is dropped (the resulting gap is not interpolated), and static full-period connectedness is estimated separately for 2011–2019 and 2021–2026.

5 Results

5.1 Full-sample networks

Table 4 (appendix) reports static connectedness estimated on the whole sample. The two dense estimators give almost the same total connectedness (Diebold–Yilmaz 81.9%, lasso VAR 82.1%), NETS gives 73.5%, and the idiosyncratic component identified by FNETS gives only 34.2%. The heatmap of Figure 3(a) shows that Diebold–Yilmaz connectedness is spread widely across all pairs, with only mild regional clustering: the strongest pairs are Korea–Taiwan and the Indonesia–Malaysia–Philippines–Thailand block, and South Africa is a common source of shocks for almost every market. Once the common factor is removed (Figure 3(b)), the remaining pairwise shares are concentrated in a few pairs (Korea–Taiwan above all), and the full-sample idiosyncratic partial-correlation network is empty (Figure 4, right panel). NETS retains 92 long-run partial-correlation links (60% of all possible pairs). The lagged (Granger) dependence is very weak in all sparse estimators: lasso, NETS and FNETS keep only 2.5%, 1.7% and 0.3% of the possible directed lag links in the rolling windows, so connectedness at the weekly frequency is overwhelmingly contemporaneous.

5.2 Connectedness over time

Figure 1 shows the rolling total connectedness of each method. Diebold–Yilmaz connectedness moves between about 72% and 89%; the lasso VAR tracks it almost exactly (correlation 0.98). Because each estimate is dated at the end of its 104-week window, an episode affects the series for two years after it occurs. Connectedness is high at the start of the sample (88%), falls to about 76% in 2014–2015, rises to about 87% in 2016–2017, is around 80% in 2018–2019, and reaches its maximum (about 89%) for windows that contain the March 2020 crash, falling back to the mid-70s after those weeks leave the window (2022–2023). NETS follows the same low-frequency pattern at a lower level (about 45–83%) but is much noisier from one window to the

next. FNETS idiosyncratic connectedness is flatter (mostly 43–55%) and does not reproduce the 2020 peak. Figure 2 shows why: the share of variance that FNETS attributes to the common component, which is between 47% and 75%, rises and falls with total connectedness, with a correlation of 0.95.

5.3 Net transmitters and receivers

Figure 5 reports average net connectedness by country. In the Diebold–Yilmaz and lasso systems the main net transmitters are South Africa (+22 percentage points), South Korea (+16) and Taiwan (+12), and the main net receivers are Turkey (−22), Vietnam (−21) and Argentina (−10). NETS gives the same ranking at about half the magnitude (South Africa +10, Korea +9, Taiwan +8; Vietnam −11, Turkey −11, Argentina −5). The ranking of the countries' average net position is almost identical across DY, lasso and NETS (Spearman correlation between 0.96 and 0.99) but differs for FNETS (0.30 with DY). In the idiosyncratic network Taiwan (+9) and Indonesia (+8) are the main transmitters, Turkey is still the main receiver (−6), and Argentina is close to neutral (+1.6). Argentina is thus a receiver of the common component of emerging-market returns rather than of idiosyncratic shocks; in this sense it is one case among the 17 markets rather than an outlier.

5.4 Comparison of methods

Table 2 summarizes the three criteria.

Density. The Diebold–Yilmaz VAR is dense by construction. The sparse estimators remove almost all lagged links, and NETS and FNETS also sparsify the contemporaneous structure: the share of non-zero long-run partial correlations is 100% for DY and lasso (which are implied by a full innovation covariance), 52% for NETS and 7% for FNETS. Among the generalized variance shares themselves, 35% (DY) and 34% (lasso) exceed the equal-share benchmark $1/N$, against 16% for both NETS and FNETS.

Agreement. Total connectedness is highly correlated between DY and the lasso VAR (0.98) and between DY and NETS (0.89), but not at all between DY and FNETS (0.00); the correlations of FNETS with lasso and NETS are 0.04 and 0.08. The same pattern holds for the matrices: the per-window Spearman correlation of the variance-decomposition entries with DY is 0.93 for lasso, 0.80 for NETS and −0.06 for FNETS, and the overlap of the strongest 10% of links (Jaccard) is 0.61, 0.51 and 0.16. Agreement is lower for the networks that are sparse by design: for long-run partial correlations the Spearman correlation with DY is 0.33 (lasso) and 0.36 (NETS), while lasso and NETS agree more with each other (0.66). The Granger networks barely agree (Spearman at most 0.23), as expected given how few lag links survive.

Stability. The standard deviation of total connectedness across windows is 5.1 (DY), 5.7 (lasso), 11.0 (NETS) and 4.6 (FNETS), but its mean absolute change from one window to the next is 0.7, 0.8, 4.6 and 3.0: the dense estimators move smoothly, while NETS and FNETS jump. The correlation between consecutive windows' variance-decomposition matrices is 0.97 (DY), 0.97 (lasso), 0.93 (NETS) and 0.82 (FNETS). For long-run partial-correlation links the overlap of the strongest 10% between consecutive windows is 0.61, 0.64, 0.72 and 0.12, so NETS has the most persistent sparse structure and the FNETS idiosyncratic network is the least stable. The sharp one-window drops of the FNETS series coincide with windows in which the penalized idiosyncratic VAR retains several coefficients (correlation of −0.72 between total connectedness and the number of non-zero coefficients); we therefore read them as estimation noise in short windows and not as events.

Table 2: Comparison criteria across methods (179 windows, $w = 104$, $H = 4$). Densities in %; agreement measured against DY; stability is the mean Spearman correlation (variance decomposition) and top-10% Jaccard similarity (long-run partial correlations) between consecutive windows.

Method	Total conn. (%)	Granger dens. (%)	LR p.corr. dens. (%)	Corr. total w/ DY	Spearman GFEVD w/ DY	Stab. GFEVD	Stab. LRPC top
DY	81.0	100.0	100.0	–	–	0.97	0.61
Lasso VAR	80.0	2.5	100.0	0.98	0.93	0.97	0.64
NETS	61.5	1.7	51.9	0.89	0.80	0.93	0.72
FNETS (idio.)	47.2	0.3	7.3	0.00	-0.06	0.82	0.12

5.5 Common factor versus idiosyncratic connectedness

FNETS selects one common factor in every window, which explains on average 58% of the variance of the series. Total connectedness of the idiosyncratic VAR is 41% lower than that of the DY VAR on average (median 42%; between 26% and 61% across windows). The loss is stable across specifications (Table 3): 38% with 78-week windows, 43% with 130-week windows, 42% without SPY and 41% when windows that touch 2020 are removed, and it does not depend on the horizon. In the subsamples (Table 4), DY connectedness is 83.0% in 2011–2019 and 73.5% in 2021–2026, NETS falls from 69.6% to 56.0%, while idiosyncratic FNETS connectedness rises slightly (34.1% to 39.6%) as the common-factor variance share falls from 58% to 47%. A large part of the variation of DY connectedness over time is therefore variation in the weight of the common factor.

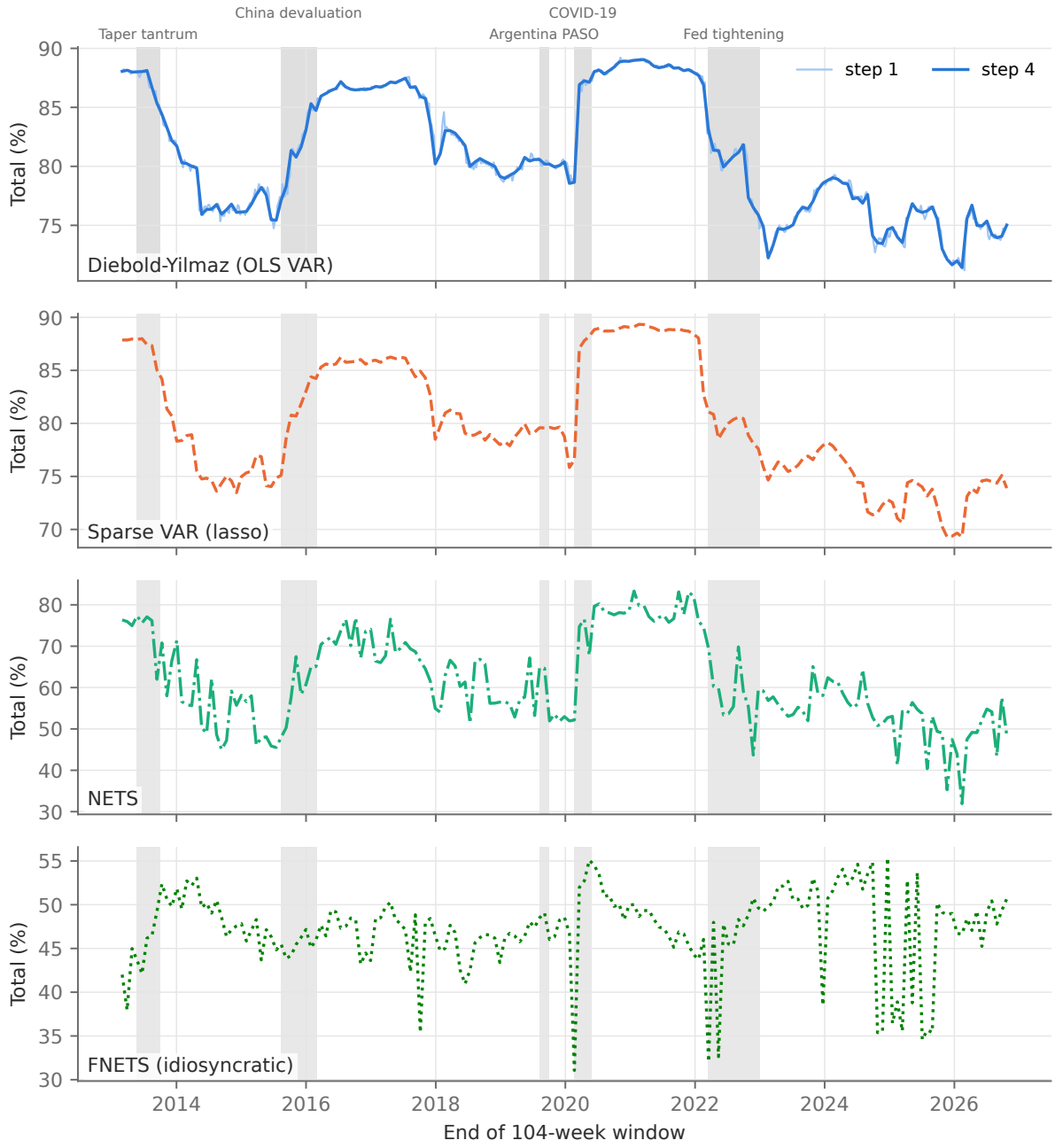


Figure 1: Total connectedness over time by method (104-week rolling windows, $H = 4$, step 4). Estimates are dated at the end of the window. Shaded areas mark events.

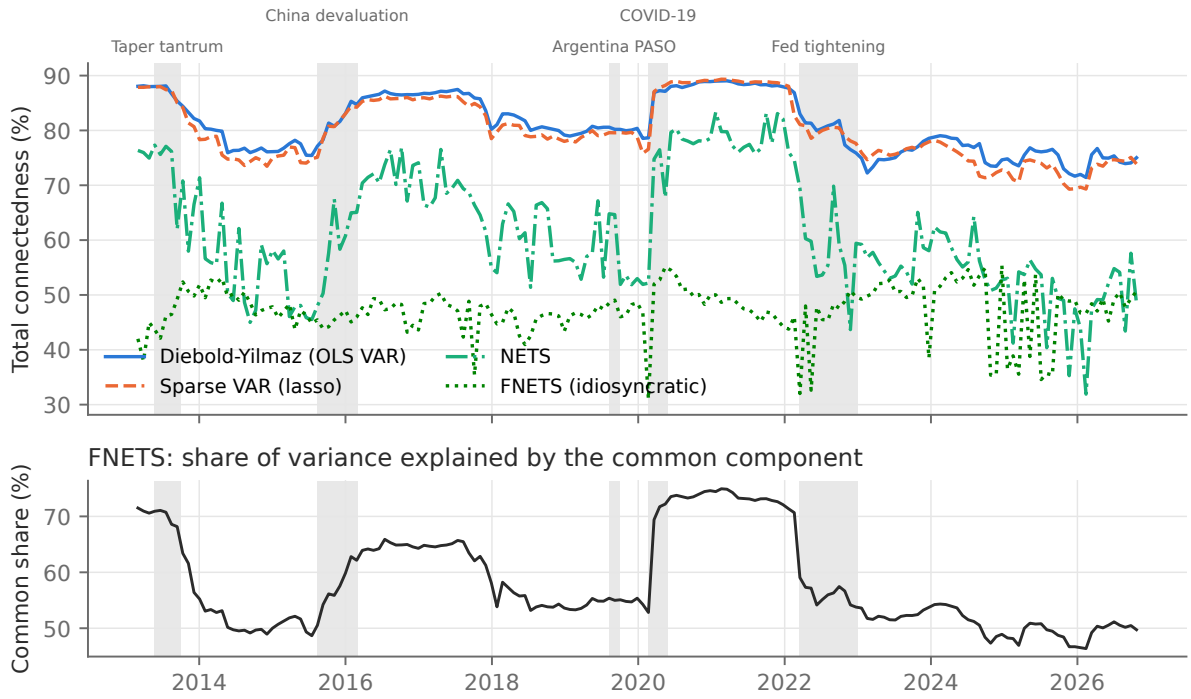
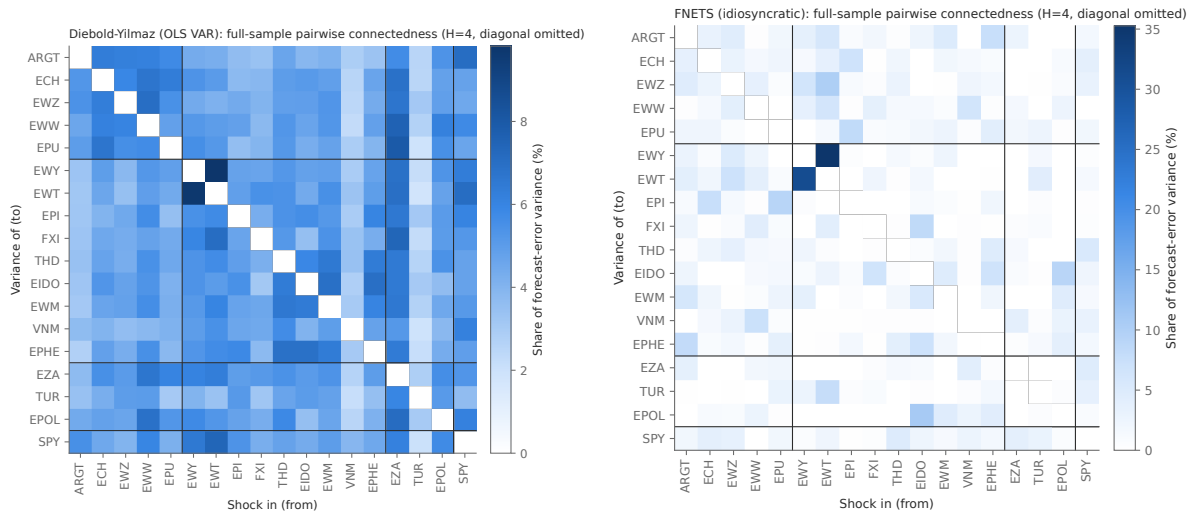


Figure 2: Top: total connectedness of all methods on a common scale. Bottom: FNETS share of variance explained by the common component.



(a) DY

(b) FNETS (idiosyncratic)

Figure 3: Full-sample pairwise connectedness, countries grouped by region (Latin America, Asia, EMEA, global).

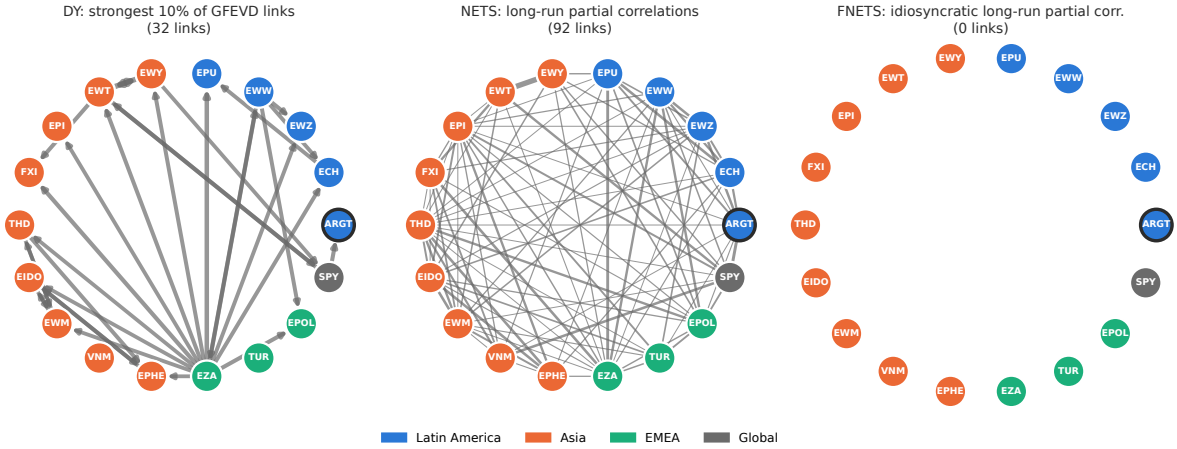


Figure 4: Full-sample networks. Node colors denote regions; ARGT is outlined. Left: strongest 10% of Diebold–Yilmaz links. Center: NETS long-run partial correlations. Right: FNETS idiosyncratic long-run partial correlations (no links in the full sample).

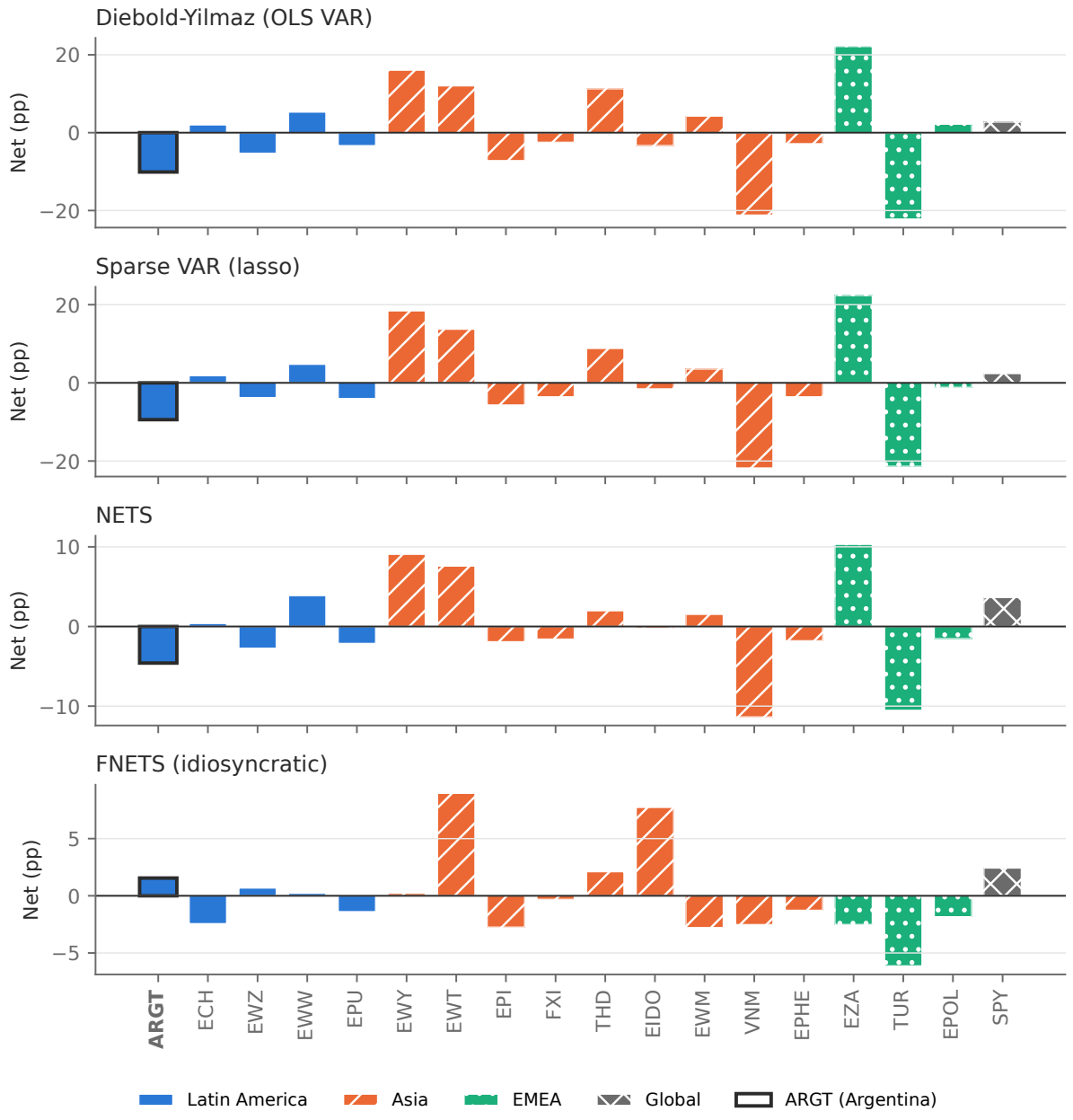


Figure 5: Average net connectedness by country (transmitters > 0, receivers < 0), in percentage points. ARGT outlined.

6 Discussion

[To be completed: Write the discussion yourselves. Evidence available in results/tables/ and docs/summary.md: (1) share of DY connectedness explained by the common factor (41%, stable across robustness) and its link with the time pattern; (2) how the method changes the picture (density, agreement, stability, Table 2); (3) interpretation of net transmitters/receivers, including Argentina (role in the paper still to be defined by the group); (4) limitations: returns instead of volatilities, ETF/currency effects, short windows for FNETS and NETS (numerical safeguards, tuning-driven drops), sparse methods leave almost no lagged links; (5) the team's database-based extension (to be added once the code is shared).]

7 Conclusion

[To be completed: Answer the research question and state what can be claimed with confidence. Final wording of the question and hypotheses is still to be decided by the group.]

8 References

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9 Data and Code Availability

The data (weekly ETF returns) and the full code are available at [repository URL]. The pipeline is written in Python; NETS and FNETS are estimated in R (packages `nets` 0.9.1 and `fnets` 0.1.6) through a command-line bridge. All parameters are in `config.yaml` and the whole analysis is reproduced with `make all` (or `python run_all.py`).

A Robustness

Table 3: Robustness: mean total connectedness (%) by method, share of DY connectedness lost once the common factor is removed (FNETS), and FNETS common-variance share.

Scenario	Windows	DY	Lasso VAR	NETS	FNETS (idio.)	Lost w/o factor (%)	Common var. (%)
Baseline (w=104, H=4)	179	81.0	80.0	61.5	47.2	41	58
H=2	179	80.9	80.0	61.5	47.2	41	58
H=8	179	81.0	80.1	61.5	47.2	41	58
w=78	185	81.2	79.6	57.8	50.5	38	58
w=130	172	81.1	80.5	63.4	46.2	43	58
Excluding SPY	179	80.0	78.9	59.9	45.9	42	58
Excluding windows touching 2020	141	79.7	78.6	58.8	47.2	41	55

Table 4: Static full-period connectedness: full sample and the subsamples before and after 2020.

Sample	Method	Total conn. (%)	LR p.corr. dens. (%)
Full sample	DY	81.9	100.0
Full sample	Lasso VAR	82.1	100.0
Full sample	NETS	73.5	60.1
Full sample	FNETS (idio.)	34.2	0.0
2011-2019	DY	83.0	100.0
2011-2019	Lasso VAR	82.8	100.0
2011-2019	NETS	69.6	60.1
2011-2019	FNETS (idio.)	34.1	2.0
2021-2026	DY	73.5	100.0
2021-2026	Lasso VAR	73.7	100.0
2021-2026	NETS	56.0	52.3
2021-2026	FNETS (idio.)	39.6	13.7

[To be completed: Comment on robustness. Not available: realized-volatility connectedness (no daily data) and local indices instead of ETFs.]

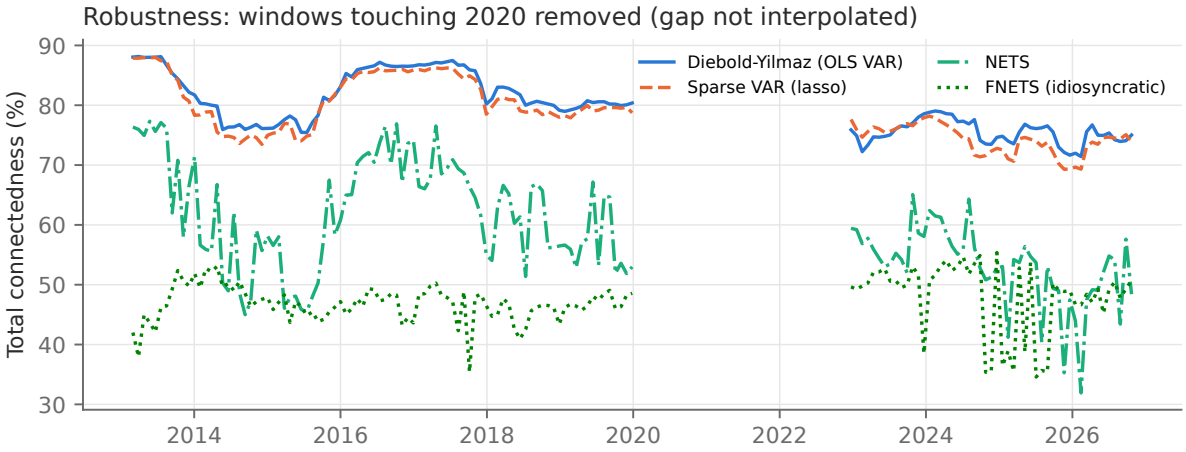


Figure 6: Total connectedness excluding every window that contains a week of 2020.